

CENG381 Stochastic Processes

Finite Markov Chains — Practice 1

Q1 (30 points). A meteorologist models daily weather with states: Sunny (S), Cloudy (C), Rainy (R). The one-step transition matrix is:

	S	C	R
S	[? 0.3 0.2]		
C	[0.4 ? 0.3]		
R	[0.1 0.5 ?]		

- Determine each missing entry so that P is a valid stochastic matrix (every row sums to 1).
- Compute the two-step transition matrix $P^{(2)} = P \cdot P$.
- Using the Chapman-Kolmogorov equation, find the three-step transition probability $p_{\{SR\}}^{(3)}$ — the probability of going from Sunny to Rainy in exactly three days.

Q2 (35 points). Consider the Markov chain on states $\{1, 2, 3, 4, 5\}$ with transition matrix:

	1	2	3	4	5
1	[1/2	1/2	0	0	0]
2	[1/2	1/2	0	0	0]
3	[0	1/3	1/3	1/3	0]
4	[0	0	0	1/3	2/3]
5	[0	0	0	2/3	1/3]

- Draw the transition diagram.
- Identify all communicating classes. State whether each class is positive recurrent, null recurrent, or transient.
- Find the limiting state probabilities. (If there are multiple recurrent classes, give the limiting distribution restricted to each class, and state the absorption probabilities for the transient state.)

Q3 (35 points). A student's daily mood follows a Markov chain on $\{\text{Good (G), Neutral (N), Bad (B)}\}$:

From G :	stay G	0.6,	go N	0.3,	go B	0.1
From N :	go G	0.4,	stay N	0.3,	go B	0.3
From B :	go G	0.2,	go N	0.5,	stay B	0.3

- Write the 3×3 transition matrix P with rows and columns ordered (G, N, B).

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- b) The student is Neutral on Monday. Find the probability distribution over {G, N, B} on Wednesday by computing $\mu_1 = (0, 1, 0) \cdot P$ and $\mu_2 = \mu_1 \cdot P$.
- c) Use the Chapman-Kolmogorov equation to find $p_{\{GB\}}^{(3)}$ — the probability of going from Good to Bad in exactly three days. First compute row G of $P^{(2)}$, then apply $p_{\{GB\}}^{(3)} = \sum_k p_{\{Gk\}}^{(2)} \cdot P_{\{kB\}}$.